Information Retrieval
Programming Assignment III
Learning to Rank
Submission deadline: November 6

General Guidelines: The project is to be done in a group. ONLY ONE MEMBER PER GROUP SHOULD SUBMIT THE ASSIGNMENT IN MOODLE. Plagiarism will be penalised. Marks will be awarded depending on completeness and correctness. You need to submit your programs as well as a writeup on your implementation details, plan of experiments and results. You may use any programming language, but python is preferable.

Text Collection: You will use the Cranfield text collection. The file cran.tar.gz has all the relevant data and the relevance judgments to be used for machine learning.

Problem Description:
The assignment is about using *machine learning* algorithms to rank documents. In the assignment you will perform ranked retrieval on the Cranfield text collection and evaluate the retrieval performance on the test queries and the human relevance judgments provided for these queries. Retrieval performance is to be measured using precision, recall, and F1 score. The following two methods should be used to learn the rankings:

Method A: Boolean Weighted Zone Scoring
Write a program to learn the weights of the individual fields of a document present in the Cranfield collection. The fields are: title, author, bibliography, and text body. Individual Boolean matching score of each of these fields with the query may be considered as input. Relevance judgments may be used as output for the machine learning problem. A grid search algorithm can be used to learn the based weights. The weights should be used to compute the final ranking score. Output should be the ranked documents relevant to a query. Evaluation should be performed for this result list.

Method B: Learning Real Valued Weights using Support Vector Machines
In this method, the real valued weights should be learned using a linear SVM using the following input features for a query document pair as well as the output relevance judgment. The learned weights may be used to obtain the final ranked list of relevant documents for a test subset of query. Precision, recall, F1 score may be used as evaluation measures.

Features:
*query term number, query term ratio, stream length,idf, sum of term, frequency, min of term frequency, max of term frequency, mean of term, frequency, variance of term frequency, sum of stream length normalized term, frequency, min of stream length normalized term frequency, max of stream, length normalized term frequency, mean of stream length normalized term, frequency, variance of stream length normalized term frequency, sum of tf*idf,, min of tf*idf, max of tf*idf, mean of tf*idf, variance of tf*idf, boolean model,, vector space model, BM25,*

The terrier search engine, or any other NLP tool may be used to compute the above features if necessary. SVM libraries may be used for machine learning.

Submissions: All the program files, result files for the test queries as a text file, and a report as a pdf file. The report may be about 10 pages long. The report should contain:
1. Group details
2. Problem statement
3. Implementation details
4. Plan of experiments
5. Results
6. Discussion of results

The submitted files will be evaluated on a further set of unknown test queries using an evaluation script. The top three groups with the best performance will receive bonus points.